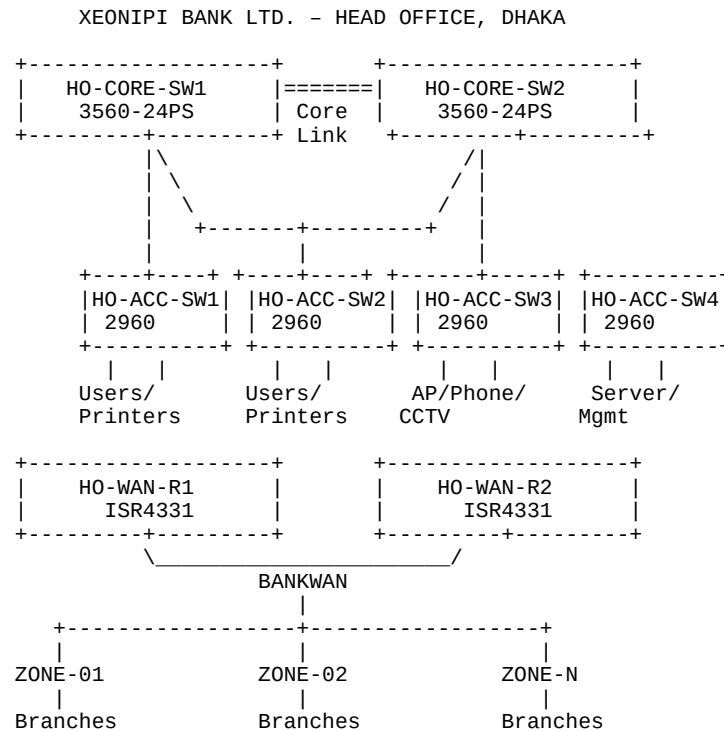


Xeonipi Bank Ltd. – BankWAN Phase 1

Cisco Packet Tracer Design Blueprint – Project I (Design Phase)



HO VLAN plan

VLAN	Name	Primary use
10	TELLER	Teller/cash workstations and local printer
20	ACCOUNTS	Accounts/loan workstations and local printer
30	HR	HR workstations and local printer
40	IT	IT/admin workstations
50	GUEST	Guest wireless clients via AP
60	VOICE	Cisco IP phones
70	CCTV	IP cameras / surveillance devices

80	SERVER	Enterprise servers and services
90	MANAGEMENT	Network administration endpoints

Recommended HO Port Mapping – Physical Design

These are design-stage port assignments. Configuration will be performed only after the physical topology is verified.

Device A	Port A	Device B	Port B	Purpose
HO-CORE-SW1	Gi0/1	HO-CORE-SW2	Gi0/1	Core interconnect
HO-CORE-SW1	Gi0/2	HO-ACC-SW1	Gi0/1	Trunk/uplink
HO-CORE-SW1	Gi0/3	HO-ACC-SW2	Gi0/1	Trunk/uplink
HO-CORE-SW1	Gi0/4	HO-ACC-SW3	Gi0/1	Trunk/uplink
HO-CORE-SW1	Gi0/5	HO-ACC-SW4	Gi0/1	Trunk/uplink
HO-CORE-SW2	Gi0/2	HO-ACC-SW1	Gi0/2	Redundant uplink
HO-CORE-SW2	Gi0/3	HO-ACC-SW2	Gi0/2	Redundant uplink
HO-CORE-SW2	Gi0/4	HO-ACC-SW3	Gi0/2	Redundant uplink
HO-CORE-SW2	Gi0/5	HO-ACC-SW4	Gi0/2	Redundant uplink
HO-CORE-SW1	Gi0/6	HO-WAN-R1	Gi0/0/0	Primary WAN edge
HO-CORE-SW2	Gi0/6	HO-WAN-R2	Gi0/0/0	Secondary WAN edge

Access-switch endpoint allocation

Switch	VLANs	Suggested endpoint groups
HO-ACC-SW1	10, 20	Teller PCs/printer/phone; Accounts PCs/printer/phone
HO-ACC-SW2	30, 40	HR PCs/printer/phone; IT PCs/laptops/printer
HO-ACC-SW3	50, 60, 70	Guest AP; IP phones; CCTV cameras
HO-ACC-SW4	80, 90	Servers; management/admin PCs/laptops

Design Rules for Zone and Branch Expansion

- HO controls all Zone Offices through the BankWAN WAN/routing hierarchy.
- Each Zone controls its subordinate Branch Offices and provides regional/local services as defined by the project.
- Branch offices do not need direct Internet access in the initial design; traffic can be carried through the Zone/HO hierarchy.
- The same Zone and Branch templates can be replicated without fixing a maximum number of Zones or Branches.
- Project I will use Cisco Packet Tracer for topology and network-service simulation; Project II will use Windows Server 2022 for the real enterprise-service lab.
- Security in Project I will emphasize VLAN segmentation, ACLs, port security, STP protections and routing controls; no ASA/Sophos firewall is required for the approved Packet Tracer scope.